

Ex 7 Random Sampling and Sampling Distribution

```
import numpy as np
import matplotlib.pyplot as plt

# Population
population = np.arange(1, 61)

# Random sampling
sample_size = 10
sample = np.random.choice(population, sample_size, replace=False)

print("Population:", population)
print("Random Sample:", sample)
print("Sample Mean:", np.mean(sample))

# Sampling distribution
sample_means = []

for i in range(1000):
    sample = np.random.choice(population, sample_size, replace=False)
    sample_means.append(np.mean(sample))

# Display sampling distribution
plt.hist(sample_means, bins=20, edgecolor='black')
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.title("Random Sampling and Sampling Distribution")
plt.show()
```

```
Population: [ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
...         25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
...         49 50 51 52 53 54 55 56 57 58 59 60]
Random Sample: [ 9 12  4 27 17 39 19 25 59 20]
Sample Mean: 23.1
```

